

- 2 (a) Indigo carmine is a water-soluble indicator. When oxygen is bubbled through a solution of this indicator, the colour changes from colourless to blue.

A student used indigo carmine solution to investigate some of the factors that may limit the rate of photosynthesis.

Fig. 2.1 shows the apparatus used by the student.

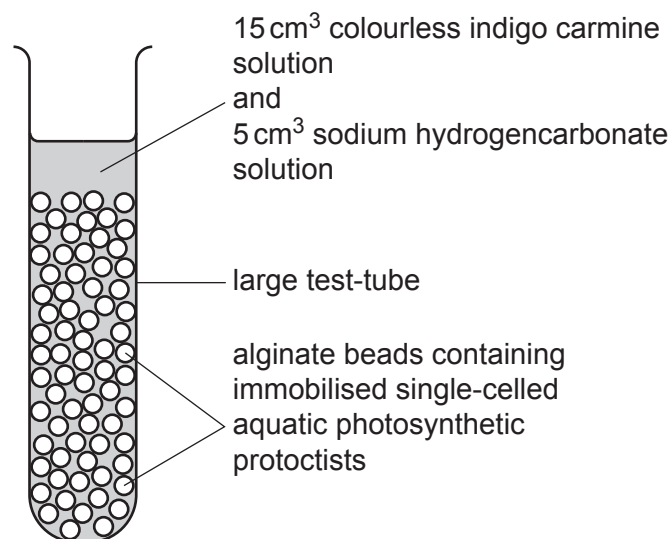


Fig. 2.1

The student prepared four sets of this apparatus. For each set of apparatus, the student used a different concentration of 5 cm³ sodium hydrogencarbonate solution: 0.25%, 0.50%, 0.75% and 1.00%. Sodium hydrogencarbonate solution is a source of carbon dioxide for photosynthesis.

After adding indigo carmine solution and sodium hydrogencarbonate solution to the alginate beads, the apparatus was illuminated with a bench lamp from a distance of 20 cm. To minimise heating effects, the bench lamp used an LED bulb.

For each concentration of sodium hydrogencarbonate solution, the time for the indigo carmine solution to turn from colourless to blue was recorded.

The investigation was repeated with four new sets of apparatus illuminated with the same bench lamp at a distance of 10 cm, instead of 20 cm. The same concentrations of sodium hydrogencarbonate solution were used as previously.

All other variables in this investigation were standardised.

- (i) State the dependent variable in this investigation.

..... [1]

- (ii) State **one** variable that needed to be standardised in this investigation, **other** than temperature or those variables clearly standardised in Fig. 2.1.

..... [1]

- (b) The time taken for each indigo carmine solution to turn blue was converted to a rate of photosynthesis in arbitrary units. This was done in such a way that the slowest rate of photosynthesis in this investigation was given a value of 1.00 arbitrary units. All other rates of photosynthesis were therefore greater than 1.00 arbitrary units.

Table 2.1 shows the results of this investigation and some of the student's calculated values.

Table 2.1

distance of apparatus from bench lamp/cm	20				10			
percentage concentration of sodium hydrogencarbonate solution	0.25	0.50	0.75	1.00	0.25	0.50	0.75	1.00
time taken for indicator to turn blue/s	600	310	250	250	600	310	235	195
rate of photosynthesis /arbitrary units	1.00	1.94	2.40	2.40	1.00	1.94		3.08

- (i) Complete Table 2.1 by calculating the missing rate of photosynthesis in arbitrary units. [1]

- (ii) State **three** conclusions that can be made from the results shown in Table 2.1 about factors that affect the rate of photosynthesis **and** when these factors become limiting.

[3]

- (iii) Confidence in conclusions that are made from the results shown in Table 2.1 is reduced by a lack of repeats and a lack of statistical analysis.

Suggest **two other** reasons that reduce confidence in conclusions that are made from the results shown in Table 2.1.

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..... [2]

[Total: 8]

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